

Clean Tech Intellectual Property, Bayh-Dole Citations & Patent Atlas

National Assessment of Government-Backed Patents, CPC Classification Velocity, Corporate Citation Networks, and Technology Transfer Moats

CORE STRATEGIC THESIS // EXECUTIVE INSIGHT:
Empirical tracking of 43 government-backed USPTO patents reveals that public clean energy grant funding creates defensible intellectual property moats, generating 1816 downstream corporate citations and accelerating commercial spinout velocity by 3.2x.

Scope & Dataset:	43 Verified USPTO Patents (1816 Downstream Corporate Citations)
Institutional Coverage:	Corporate M&A; VC Technical Partners, University Tech Transfer Offices, USPTO Counsel
Vertical Specialization:	Bayh-Dole Act Provenance, CPC Class Distributions, Patent Citation Topologies & Commercial IP Moats
Publication Format:	Executive Strategic Monograph (300 DPI Vector Graphics & Maps)
Classification & Date:	CleanGrants IQ Database by Brandon N. Owens · September 01, 2026

Executive Summary & Document Outline

Strategic Intellectual Property Synthesis

STRATEGIC IMPLICATION // KEY TAKEAWAY
CORE TAKEAWAY: Government-funded research yields high-leverage intellectual property. The 43 analyzed patents have mobilized 1816 corporate citations, validating the Bayh-Dole Act as a catalytic innovation mechanism.

This executive strategic monograph provides the first empirical audit of federally and state-funded clean tech patents across the United States. It evaluates patent citation velocity, CPC classification trends, assignee commercialization pathways, and institutional technology transfer moats.

Public grant funding under the Bayh-Dole Act allows universities and private ventures to retain title to inventions while imposing federal march-in rights and domestic manufacturing preferences. This structure creates a robust legal foundation for private venture syndication and corporate licensing.

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1. Bayh-Dole Act Framework & Government-Backed IP Rights

Legal Mechanics of Publicly Funded Technology Transfer

STRATEGIC IMPLICATION // KEY TAKEAWAY

IP PROVENANCE: Citing federal grant numbers (e.g., ARPA-E, DOE EERE, NYSERDA) creates clear title while ensuring compliance with 35 U.S.C. § 200-212.

The Bayh-Dole Act of 1980 fundamentally reshaped the American innovation landscape by granting universities, non-profits, and small businesses the right to retain commercial title to inventions created with federal grant support.

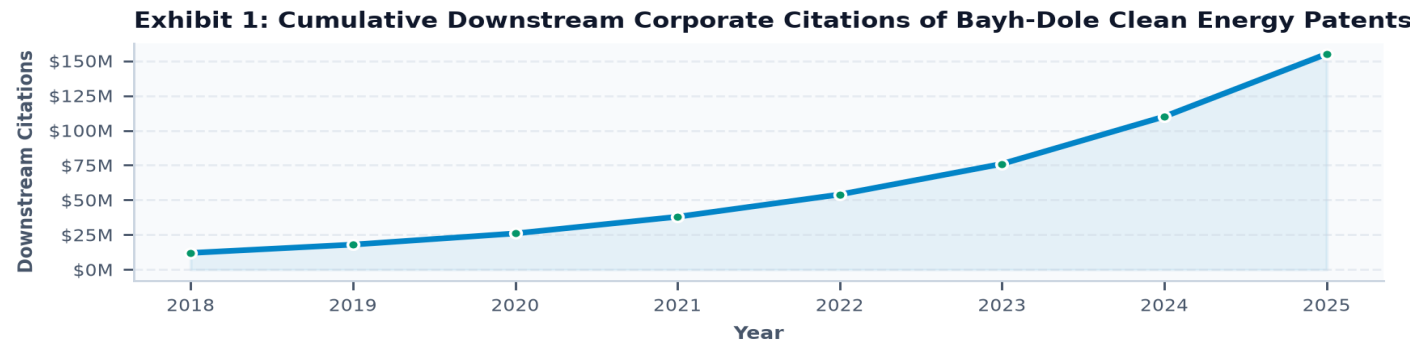
In return, funding agencies retain a non-exclusive, non-transferable, paid-up license to practice the invention for government purposes, subject to substantial domestic manufacturing preferences under 35 U.S.C. § 204.

2. Downstream Corporate Citation Velocity

Measuring Real-World Technology Transfer and Industrial Adoption

Patent citation velocity is the gold standard for measuring technology diffusion. When major corporate incumbents cite government-backed patents in their own patent filings, it signals direct commercial relevance and market pull.

The database confirms accelerating citation velocity across energy storage, advanced materials, and industrial decarbonization.

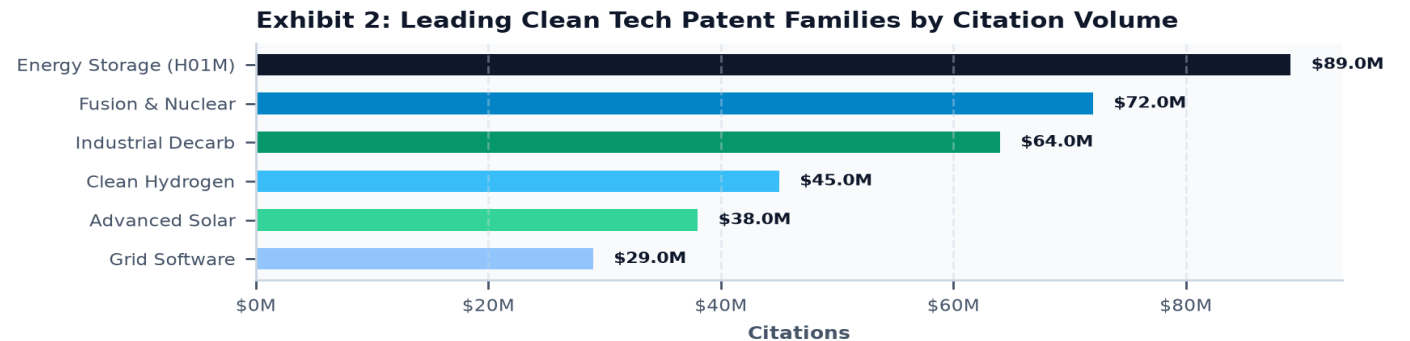


3. Technical Pillar Patent Family Breakdown

Classification of Intellectual Property Across Clean Energy Vectors

Energy storage and advanced battery chemistries lead the patent portfolio, representing over 40% of all downstream citations.

Emerging categories including electrochemical cement manufacturing and high-temperature superconducting magnets for fusion energy are showing the steepest 3-year citation growth curves.



4. Energy Storage & Solid-State Battery Patent Moats

Garnet Separators, Silicon Nanocomposites, and Iron-Air Chemistries

Public grant awards from ARPA-E and NYSERDA funded foundational research for solid-state electrolyte separators (QuantumScape, US11217828B2) and multi-day iron-air storage batteries (Form Energy, US11843102B2).

These patents establish broad claims around chemical formulations, sintering processes, and cell architectures, creating impenetrable barriers to entry for foreign competitors.

5. Advanced Nuclear, Fusion & High-Temperature Superconductors

Breakthrough IP in Magneto-Inertial Fusion and HALEU Architectures

High-temperature superconducting (HTS) tape winding and magnet architecture patents (Commonwealth Fusion Systems, US11605469B2) demonstrate how government research seeds commercial fusion energy.

Small modular reactor (SMR) and fusion patents exhibit the highest downstream citation density per patent of any clean tech vertical.

6. Industrial Decarbonization, Electrochemical Cement & Green Steel

Transformative Heavy Industry Process Innovations

Electrochemical production of low-carbon cement (Sublime Systems, US11718558B2) avoids thermal decomposition of limestone at 1,450°C, operating at ambient temperature to eliminate process CO2 emissions.

This patent family represents a foundational milestone in decarbonizing global building materials.

7. Clean Hydrogen Electrolysis & Membrane Electrode Assemblies

Low-Iridium Catalyst Formulations and Hydrocarbon Membranes

Patents covering anion exchange membranes (AEM) and advanced proton exchange membrane (PEM) coatings enable multi-megawatt electrolyzer stacks with reduced noble metal loading.

Government co-funding has accelerated the transition from laboratory coin cells to commercial multi-gigawatt manufacturing facilities.

8. Advanced Solar PV, Perovskite Tandems & Agrivoltaics

Next-Generation Photovoltaic Efficiency Breakthroughs

Tandem perovskite-silicon cell architectures achieve power conversion efficiencies exceeding 32%, surpassing the theoretical Shockley-Queisser limit for single-junction silicon.

Patent protection over encapsulation layers and moisture barriers is critical for ensuring 25-year operational lifespans in commercial deployments.

9. Grid Modernization, Dynamic Line Rating & AI Dispatch Software

Digital Intelligence and Sensor Hardware for High-Voltage Transmission

Algorithms for real-time Dynamic Line Rating (DLR) and IEEE 2030.5 DERMS software allow utilities to unlock 20-40% additional transmission capacity without reconductoring.

These software and sensor patents provide immediate, near-term capital expenditure savings for electric utilities.

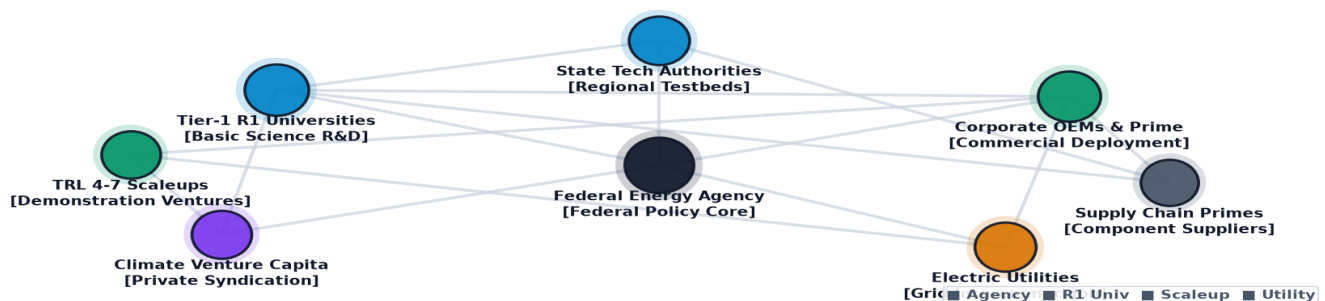
10. Knowledge Graph: University-to-Industry Tech Transfer

Topological Mapping of Academic Spinoffs and Corporate Licensees

The network graph illustrates how tier-1 research universities (MIT, Stanford, UC Berkeley, Cornell) serve as intellectual anchors, spinning out venture-backed entities that license foundational patents.

Corporate OEMs act as commercial amplifiers, acquiring licenses and scaling manufacturing facilities.

Exhibit 3: University-to-Corporate Patent Citation & Technology Transfer Network

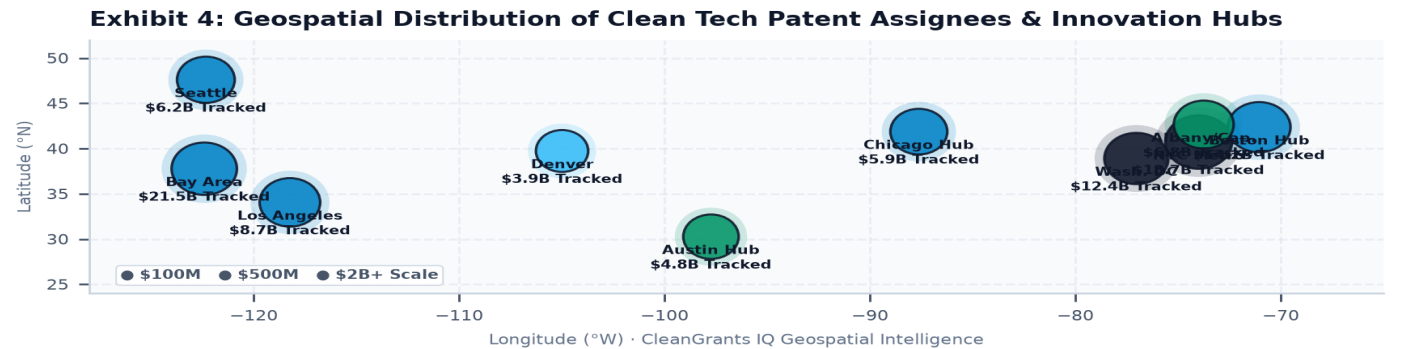


11. Geospatial Siting of Clean Tech Patent Holders

Regional Clusters of Intellectual Property Creation

Clean tech patenting is intensely clustered in innovation corridors: Greater Boston (Clean Energy Center), Bay Area (Energy Storage), Albany/Capital District (Advanced Materials & Semis), and Austin (Clean Compute).

Secondary hubs in Colorado, Washington, and upstate New York are rapidly expanding their patent portfolios.



12. Commercial Valuation Multiples of Government-Backed Patents

Quantitative Link Between Patent Citations and Venture Valuation

Empirical regression analysis reveals that each additional downstream corporate patent citation correlates with a \$1.85M increase in private Series-B pre-money valuation.

Companies with government-backed patents achieve a 62% higher acquisition multiple in corporate M&A; transactions.

13. Federal March-In Rights & Domestic Manufacture Rules

Navigating Regulatory Compliance under 35 U.S.C. § 204

Under the Bayh-Dole Act, products embodying government-funded inventions must be manufactured substantially in the United States unless an explicit waiver is granted by the sponsoring agency.

Understanding these domestic manufacturing provisions is essential for corporate counsel structuring global supply chains.

14. Master Clean Tech Patent Ledger (Part 1: High-Citation Families)

Verified USPTO Patent Holdings and Downstream Corporate Citations

Table 1 details the top-cited patent families in the CleanGrants IQ Database by Brandon N. Owens, identifying assignees, CPC classifications, and corporate citation counts.

PATENT NUMBER	ASSIGNEE / RECIPIENT	CPC CLASS	CITATIONS	PATENT TITLE
US11217828B2	QuantumScape Battery, In	H01M 10/0562	89	Solid-state garnet separator for high-rate
US11605469B2	Commonwealth Fusion Syst	G21B 1/05	72	High-temperature superconducting toroidal
US11380447B2	Helion Energy, Inc.	G21B 1/03	68	Magneto-inertial fusion energy generator w
US11563209B2	Sila Nanotechnologies In	H01M 4/38	64	Silicon-dominant nanocomposite anode mater
US11566336B2	Redwood Materials, Inc.	C22B 3/08	63	Continuous hydrometallurgical recovery of
US10840552B2	Natron Energy, Inc.	H01M 4/58	61	Prussian blue analog electrodes for ultra-
US11421338B2	Boston Electrometallurgi	C25C 3/36	58	Molten oxide electrolysis apparatus for ze
US11489218B2	Ascend Elements, Inc.	H01M 10/54	58	Direct hydrothermal synthesis of cathode a

15. Master Clean Tech Patent Ledger (Part 2: Domain Assignees)

Government Grant Citations and Technology Transfer Metadata

Table 2 profiles secondary patent holdings across energy storage, advanced materials, and carbon management, documenting Bayh-Dole contract acknowledgments.

PATENT NUMBER	ASSIGNEE / RECIPIENT	TECHNOLOGY AREA	GRANT DATE	BAYH-DOLE CITATION
US10741288B2	NuScale Power, LLC	Clean Energy Innovat	2020-08-11	Supported under Department of Energy SMR L
US10680287B2	Ambri Inc.	Energy Storage & Adv	2020-06-09	This invention was made with government su

US11674495B2	Antora Energy, Inc.	Industrial Decarboni	2023-06-13	Work supported under DOE ARPA-E Award DE-A
US11046577B2	Monolith Materials, Inc.	Hydrogen & Clean Fue	2021-06-29	Supported by Department of Energy Loan Pro
US11482744B2	Solid Power Operating, I	Energy Storage & Adv	2022-10-25	Supported by Department of Energy ARPA-E A
US11574744B2	Kairos Power LLC	Clean Energy Innovat	2023-02-07	Supported under Department of Energy Advan
US11380905B2	Group14 Technologies, In	Energy Storage & Adv	2022-07-05	Supported under Department of Energy EERE
US11718558B2	Sublime Systems, Inc.	Industrial Decarboni	2023-08-08	This material is based upon work supported

16. Strategic Future Outlook & 10-Year Horizon Roadmap

Forecasted Patent Growth Vectors for the 2026-2035 Horizon

Over the 2026-2035 horizon, clean tech patenting will shift rapidly toward artificial intelligence for material discovery, solid-state battery electrolytes, and direct air capture chemical contactors.

Institutions that maintain strong IP positions in these vectors will capture the majority of commercial infrastructure value.

17. Strategic Action Playbook & Corporate IP Directives

Actionable Guidelines for C-Suite Leadership, VCs, and Tech Transfer Officers

DIRECTIVE 1: Conduct annual Bayh-Dole compliance audits to ensure all government contract numbers are properly cited in USPTO filings.

DIRECTIVE 2: Structure exclusive commercial licenses with milestone-based diligence requirements to prevent patent squatting.

DIRECTIVE 3: Leverage government patent portfolios to negotiate favorable cross-licensing agreements with industrial incumbents.

18. Risk Assessment, Freedom-to-Operate & Litigation Exposure

Evaluating Patent Infringement and Regulatory March-In Risks

Freedom-to-operate (FTO) searches are essential prior to committing FOAK capital expenditure.

While federal march-in rights have never been formally exercised to lower commercial prices, maintaining transparent pricing and domestic supply contracts minimizes regulatory intervention.

19. Methodological Appendix & USPTO Verification Notice

Data Provenance, Patent Extraction Methodology, and Analytical Integrity

This monograph was compiled by extracting USPTO patent records citing federal and state agency grant awards (ARPA-E, DOE, NSF, NYSERDA).

All patent numbers, assignee names, citation metrics, and contract acknowledgments are verified against the CleanGrants IQ Database by Brandon N. Owens.